certificateSigning

Service Description

Service ID: *"certificateSigning"*

Abstract
This document describes the `certificateSigning` service, version 1.0.0, produced by the CertificateAuthority support system of Arrowhead (documentation edition 5.3.0).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `certificateSigning` service, which provides signing of proof-of-possession certificate requests with the local-cloud CA, under the Arrowhead certificate profiles. It is produced by the CertificateAuthority system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

2.1 Significant Prior Art

Successor of the Arrowhead 4.6 Certificate Authority signing capability; X.509 and PKCS\#10 after RFC 5280 and RFC 2986.

2.2 How This Service Is Meant to Be Used

Onboarding (and Sysop) sends a CSR made with the subject's own private key and receives the certificate and chain. The CA builds subject and extensions itself; nothing but the public key and requested addresses is taken from the CSR.

2.3 Important Delimitations and Dependencies

Only provisioning identities may sign, and only their profiles: Onboarding `on`, `de`, `sy`; Sysop also `op`; Certificate Authority `sy`. The common name must be `<Name>.<Cloud>.<Organization>.arrowhead.eu` of this cloud; reserved system names need Sysop; a name with a valid certificate of another issuer is refused. Onboarding certificates last at most 24 h, others 365 days.

3 Service Interface

This section describes the interfaces to the `certificateSigning` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.



Figure 1: SysML-style block of the `certificateSigning` service and its operations.

The following interface operations are available.

3.1 operation sign (CertificateSigningRequest) : CertificateSigningResult

Signs a CSR for the requester's permitted profile and records the certificate.

4 Information Model

Here, all data objects that can be part of the `certificateSigning` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.3, which are used to represent things like hashes and identifiers.

4.1 struct CertificateSigningRequest

Field	Type	Description
encodedCSR	String	PEM PKCS\#10 request or Base64 DER (alias `csr`).
profile	String	`on`, `de`, `sy` or `op`; default the CSR dnQualifier, else `sy`. Must be allowed for the caller.
addresses	List<String>	DNS names and IP addresses for the subject alternative name (at most 16); default those of the CSR.
validAfter	DateTime	Start of validity, ISO 8601; default now minus one minute.
validBefore	DateTime	End of validity; default start plus the profile maximum (24 h for `on`, 365 days otherwise), never beyond the CA certificate.

4.2 struct CertificateSigningResult

Field	Type	Description
id	Number	Record identifier in the certificate register.
certificateChain	List<String>	Issued certificate and CA certificate, Base64 DER, leaf first.
commonName	String	Subject common name.
profile	String	Granted profile: `ON_BOARDING`, `DEVICE`, `SYSTEM` or `OPERATOR`.
serialNumber	String	Serial number, decimal.
validAfter	DateTime	Start of validity.
validBefore	DateTime	End of validity.

4.3 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
DateTime	Pinpoints a specific moment in time (UTC).
List	An *array* of a known number of items, each having type A.
Number	An integer number.
String	A chain of characters.

5 References

[1] J. Delsing, “CertificateAuthority – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-CertificateAuthority-java-spring/documentation/v5.3.0/CertificateAuthority-v5.3.0-SysD.pdf.

[2] ——, “certificateSigning HTTP/TLS/JSON – Interface Design Description (CertificateAuthority),” Eclipse Arrowhead, IDD 5.3.0, 2026, certificateAuthority-v5.3.0-certificateSigning-HTTP-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)
3	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	